

Hardware IP Card

Basic Information

Licensing	Closed-source proprietary
Target FPGA or ASIC	XC7K325T, ASIC
Design Language	SystemVerilog

System-Level Features

Coupling Type	Tightly-coupled
---------------	-----------------

Data Types

ISA

Peak Ops Per Cycle

Interfaces

Supported Operations

Compiler Support	Yes, specialized compiler based on LLVM17
------------------	---

External Memory Required

Cache Or Scratchpad

Architecture

Type	CPU extension
------	---------------

Num Clock Domains	1
-------------------	---

Control Interface	ISAExtension	Yes (RV32I)
-------------------	--------------	-------------

Control Interface	Memory Mapped	No
-------------------	---------------	----

Control Interface	Protocol
-------------------	----------

Memory Interface	Interface Protocol
------------------	--------------------

Memory Interface	Hierarchy Level
------------------	-----------------

Memory Interface	Cached
------------------	--------

Memory Interface	IOMMU
------------------	-------

Interrupts

Microarchitecture

Parametrization	Parametric Num Units	No
-----------------	----------------------	----

Parametrization	Parametric Config	Yes
-----------------	-------------------	-----

Programmability	Programmable Cores	No
-----------------	--------------------	----

Programmability	ISAExtension
-----------------	--------------

Software

Compiler	Requires Specialized Compiler	Yes
----------	-------------------------------	-----

Compiler	Supported Compilers	Specialized compiler based on LLVM17
----------	---------------------	--------------------------------------

Hardware Abstraction Layer	HAL Functions	No
----------------------------	---------------	----

highLevelAPI	Has High Level API	No
--------------	--------------------	----

highLevelAPI	SDK Repository
--------------	----------------

highLevelAPI	Has Domain Specific Compiler	No
Integration		
IP Distribution	Manifest Type	No
IP Distribution	Standalone Simulation	No
IP Distribution	Standalone Simulation SW Requirements	None
IP Distribution	Integration Docs Or Examples	Yes, closed-source proprietary
IP Distribution	IP-XACT	
IP Distribution	System RDL	
Synthesis	Is Synthesizable	Yes
Synthesis	FPGA Synthesis Examples	Yes, closed-source proprietary
Synthesis	ASIC Synthesis Examples	Yes, closed-source proprietary
Simulation	Closed Source Simulation	Yes (Xcelium)
Simulation	Open Source Simulation	No
Evaluation	PPA Results Available	Yes
Physical Implementation		
ASIC Area	Num Gates	14 GE added to CV32E40S
ASIC Area	On Chip Internal Memory Size	
ASIC Area	Target Clock Frequency	500 MHz (16nm)